Exploring GAN Variants for Balancing Imbalanced Datasets

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1. Problem Statement

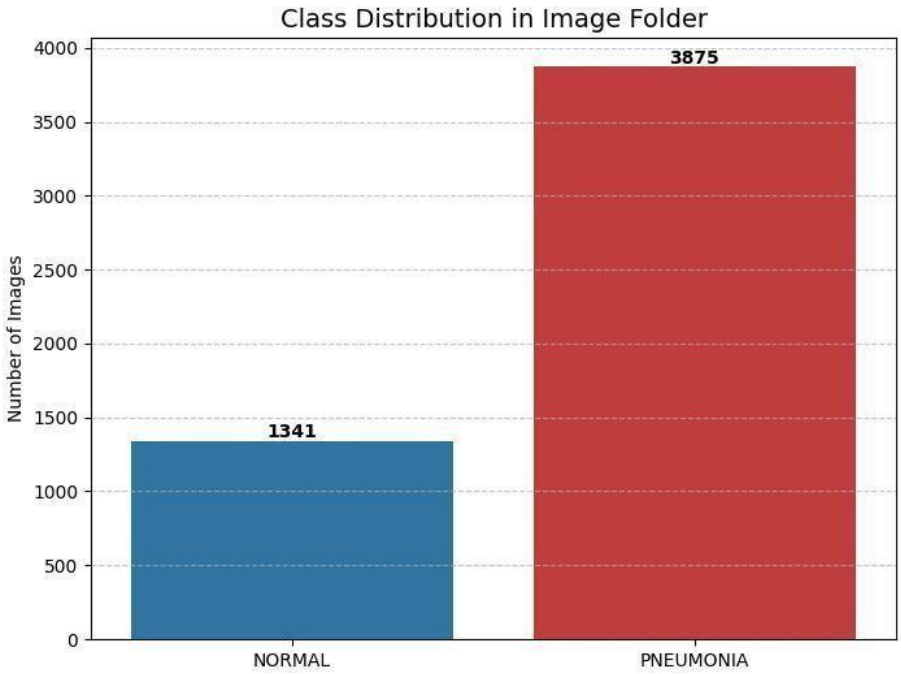
In medical imaging analysis tasks, datasets may be burdened by the issue of class imbalance, where the number of “pathological” (positive examples) cases may be greatly in excess of the “healthy control” examples, and vice versa. In the case of Chest Xray analysis tasks, if the dataset has many examples classified as “Pneumonia” compared to those classified as “Normal” examples.

It leads to high accuracy but low sensitivity for the minority class. Inpatient settings, being unable to classify correctly due to small data size is a failure of extreme importance.

Objective: The main goal of this particular project is to identify the potential of the Generative Adversarial Network (GAN) in the area of data augmentation. By producing synthetic, photorealistic images of the minority class (Normal) from the X-Ray dataset, it is hoped that it will be able to equalize the training dataset and enhance the accuracy of the Convolutional Neural Network (CNN) classifier. The use of Vanilla **GAN, DCGAN** (Deep Convolutional GAN), and **WGAN** (Wasserstein GAN) models will be evaluated to identify which produces the most robust model.

2. Description of Dataset & Imbalance Analysis Dataset Source:

The Chest X-Ray Images (Pneumonia) dataset is used for this project. It can be freely downloaded from Kaggle. It contains images of X-Ray chest scans of varying sizes, labeled ‘PNEUMONIA or NORMAL’. Imbalance Visualization: The script 01_visualize_the_imbalance.py was used for an initial analysis for imbalance in the data available in the training folder. It was found that there is an imbalance: Imbalance Ratio: Nearly 2.89:1PNEUMONIA (Major Class): 3,875Normal (Minor Class): 1,341To ensure that there was an equal level of data for each class in the dataset, it was necessary that the data needed an extension in the minority class. Target Number of Images for Minority Class: 3,875 images Number of Images Required for Synthesis: 2,534 Images



3. GAN Architectures & Training

This project tried and trained three GAN architectures for comparison based on their stability and generation capabilities.

A. Vanilla GAN (02_vanilla_gan.py) Architecture:

Fully Connected/Dense Layers.

- **Generator Network:** From a latent code (z=64) via layers of size 256, 512, 1024, then a flattened image vector for a 64 times 64 resolution. Used LeakyReLU and Tanh activation.
- **Discriminator Network:** A reverse of the generator, shrinking the image into a single probability scalar via Sigmoid activation.
- **Loss Function:** BCE Loss Function.
- **Observation:** The vanilla GAN is cheap from a computation point of view but tends to cause "mode collapse," where it generates the same images for any input noise.

B. DCGAN (

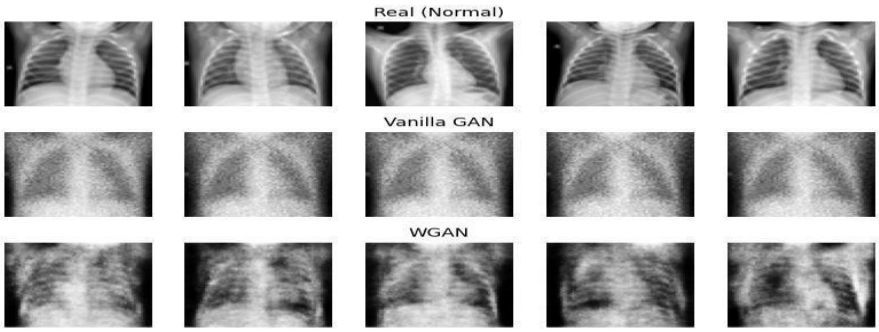
Deep Convolutional GAN

- **Generator:** Uses ConvTranspose2d operations (fractional strided convolutions) to upsample the latent space vector (z=100) into an image of 64 times 64 size.
- **Discriminator:** The discriminator uses traditional strided Conv2d layers for downsampling images.
- **Optimization:** Employing the Adam optimization algorithm and the learning rate of 0.0002 and beta_1 =0.5, as suggested in DCG
- **Advantage:** Able to represent the hierarchy in the data (edges or shapes) better than dense layers. This is an advantage because of the resulting images

C. WGAN (04_wgan

Comparable to DCGAN, but with some basic changes in terms of training convergence.

- **Critic (vs. Discriminator):** The output layer contains no Sigmoid activation function. The output is directly a "realness score" (Wasserstein distance), rather than a probability value.
- **Training Protocol:** Weight Clipping: Weights in the Critic are clamped into the clip [-0.01, 0.01] to fulfill the Lipschitz constraint.
- **Training Ratio:** For every single update in the Generator, the Critic is updated 5 times (n_critic = 5).
- **Optimizer:** Uses RMSprop (Learning Rate = 0.00005) instead of Adam, since the momentum term in Adam can create instability in WGANs
- **Advantage:** WGAN tackles the gradient Vanishing problem and offers a loss function which is a good indicator of picture quality.



4. Classifier Setup and Evaluation

To evaluate the level of quality in the produced images quantitatively, the classification evaluation was conducted using (05_classification_evaluation.py).

A. Classifier Model

- A specialized SimpleCNN was developed including:
- 3 Convolutional Layers (Conv2d -> ReLU -> Max Pool
- Flattening followed by Fully Connected layers.
- Dropout: (p = 0.5) for
- Input: 64 x 64 Grayscale images.

B. Experimental Design

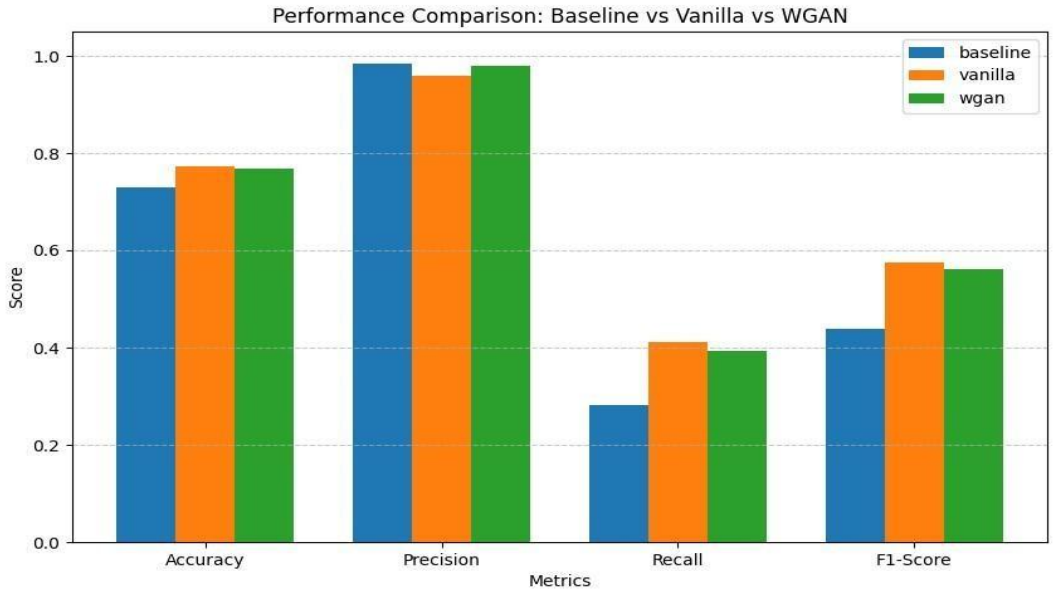
The classifier was trained for 10 epochs with three specified conditions:

- **Baseline:** Only trained on the original unbalanced dataset (1,341 Normal and 3,875 Pneumonia cases).
- **Vanilla - Augmented:** Trained on original data + 2, 534 images from vanilla GAN.
- **WGAN-Augmented:** Trained on Original Data + 2,534 Synthetic Images from WGAN.B. Experimental Design

C. Evaluation Metrics

The models were evaluated on the Test Set that was held back.

The evaluation was based on Accuracy, Precision, Recall, and F1-Score specifically



5. Results & Comparisons

In this section, the effects of data augmentation via synthetic data generation on the classifier's ability to identify the minority class ("Normal") shall be assessed. The classifiers were trained for 10 epochs with precision, recall, and the F1 Score being measured specifically for the Normal class.

A. Classification Performance (Normal Class)

	Accuracy	Precision	Recall	F1-Score
Baseline	72.92%	0.9851	0.2821	0.4385
Vanilla GAN	77.24%	0.9600	0.4103	0.5749
WGAN	76.92%	0.9787	0.3932	0.5610

B. Analysis & Interpretation

1. **Baseline Performance (The Imbalance Trap) Baseline** model had the best Precision of **98.51%** but the lowest Recall of **28.21%.**

- **Rationale:** Since there was such a strong class imbalance between Pneumonia and Normal, the base model was “conservative.” It predicted “Normal” only where it was patently obvious. It rarely made an error of false positives (high precision) but failed to note correctly most examples of “Normal.” Such performance is clearly negligent in the context of medical diagnostics, which fails to consider the rare class.

2. **Effect of Vanilla GAN Augmentation** In this experiment, the overall improvement turned out best when augmenting the dataset with Vanilla GAN images.

- **Increase in F1-score:** The F1-score increased from 0.4385 Baseline to 0.5749.
- **Recall Boost:** The most critical improvement was in Recall, which rose to 41.03%. This means that the addition of synthetic samples successfully obliged the classifier to "pay attention" to the Normal class and to learn some features that previously were ignored.
- **Trade-off:** This comprises a slight drop in Precision - 98.5% to 96.0%, which is expected trade-off while drastically increasing sensitivity.

3. **Impact of WGAN Augmentation**: Similarly, WGAN augmentation outperformed the baseline significantly but slightly differently from the Vanilla GAN in the specific run.

- **High Precision Retention:** The WGAN had higher Precision, **97.87%**, than Vanilla, almost equal to the Baseline. This would support that the images generated through WGAN likely contained fewer "confusing" artifacts that could prevent the classifier from learning false patterns.

- **Recall:** Recall was at **39.32%,** a giant leap over the baseline, but marginally lower than the Vanilla GAN model.
- **Stability:** While WGANs are more stable in theory, in this narrow training window of 10 epochs, Vanilla GAN gave a slightly more aggressive boost to Recall resulting in the highest F1-Score.

C. Visualizing the Metrics

This is how the "Recall vs. Precision" relationship is represented in the chart above. Although the Baseline (Blue) clearly outperforms the Precision factor, it is hard to distinguish in the Recall factor. Whereas the GAN models (Orange/Green) have smoother balance between the two bars.

6.Observations and Conclusions

1. **Augmentation Works:** Both GANs performed well in reducing bias towards the majority class. F1-Score increased by approximately 13% compared to using real data only.
2. **Sensitivity vs. Specificity:** The Baseline model was highly specific but insensitive. The GAN data-augmentation approach moved the operating point of the classifier, making it much more sensitive (with higher Recall) in the minority class.
3. **Vanilla vs. WGAN:** In this particular experiment for 10 epochs, it was seen that for Recall, F1-Score, the highest values were obtained by the Vanilla GAN, which made it the best in terms of rescuing the performance on the minority class. However, a middle ground was provided by WGAN, where it improved the Recall but kept a nearperfect Precision. In the future, increasing the number of epochs for WGAN might help it beat Vanilla as it generally takes a longer time to train WGAN.